MINISTRY TECHNICAL DOCUMENTATION

WCM Outreach Reporting System

Complete System Architecture, Technical Implementation, API Reference & Operations Manual

VERSION 2.0.0 — PRODUCTION READY

Organization: World Christian Ministry (WCM)
Document Scope: End-to-End System Specifications, Database Schemas & Deployment
Publication Date: October 17, 2026
Classification: Confidential Ministry Technical & Architecture Record
Repository: saikiranrayudu02-cell/wcm-outreach

WCM Outreach — Ministry Reporting System

Comprehensive System Architecture, Implementation & Operations Manual

Project Name: WCM Ministry Coordinator Reporting System ("WCM Outreach")
Organization: World Christian Ministry (WCM)
Document Version: 2.0.0
Preparation Date: October 17, 2026
Document Classification: Ministry Technical & Operational Documentation
Repository: `saikiranrayudu02-cell/wcm-outreach`

Table of Contents

1. Executive Summary
2. Project Objectives and Scope
3. Complete Feature Documentation
4. Technology Stack
5. System Architecture
6. Complete Project Folder Structure
7. User Interface Documentation
8. Database Documentation
9. API Documentation
10. Authentication and Security
11. Installation and Local Setup
12. Environment Configuration
13. Testing and Quality Assurance
14. Deployment Documentation
15. Troubleshooting Guide
16. Limitations and Future Improvements
17. User Guide
18. Developer Maintenance Guide
19. Glossary and References

1. Executive Summary

1.1 Project Purpose

The **WCM Ministry Coordinator Reporting System** (internally designated **WCM Outreach**) is an enterprise-grade ministry operations platform built for **World Christian Ministry (WCM)**. It provides a structured, verifiable, and hierarchical pipeline for reporting, reviewing, and auditing outreach meetings conducted across diverse geographical jurisdictions (States, Zones, and Districts).

1.2 Problems It Solves

Prior to the implementation of WCM Outreach, ministry data collection suffered from several critical operational challenges:

- **Fragmented Communication**: Meeting outcomes, attendance counts, and church host arrangements were communicated through unstandardized channels (messaging apps, phone calls, paper slips).
- **Delayed Decision-Making**: Leadership lacked real-time visibility into whether meetings were properly arranged, how many pastors and members attended, and what follow-up opportunities were created.
- **Lack of Accountability and Audit Trails**: Review comments and report modifications could not be tracked back to specific leaders.
- **Connectivity Barriers**: Coordinators often travel to remote rural territories with intermittent mobile connectivity, requiring offline fallback capabilities and mobile-first touch optimization.

1.3 Intended Users

The system serves four primary ministry roles:

1. **Outreach Coordinators**: Field executives on the ground who coordinate with local pastors, verify venue arrangements, record attendee statistics, and submit detailed post-meeting evaluations.
2. **Zone Leaders**: Regional overseers responsible for all coordinators and meetings within an assigned Zone (e.g., Zone-1). They edit, approve, or forward reports up the hierarchy.
3. **State Leaders**: Overseers of an entire state ministry territory (e.g., Andhra Pradesh), reviewing aggregate statistics and approving outreach activities.
4. **System Administrators**: Central ministry leadership possessing unrestricted authority over user provisioning, hierarchical delegations, global report approvals, system announcements, and ministry analytics.

---

2. Project Objectives and Scope

2.1 Primary Objectives

1. **Unified Report Lifecycle**: Consolidate Verification (pre-meeting), Attendance (meeting execution), and After-Meeting (post-meeting impact) into a single atomic report record with unique serial identification (e.g., `MR-2026-00015`).
2. **Hierarchical Multi-Tier Review**: Support report forwarding, review comments, status workflows (`DRAFT` $\to$ `SUBMITTED` $\to$ `ZONE_LEADER_REVIEW` $\to$ `STATE_LEADER_REVIEW` $\to$ `ADMIN_REVIEW` $\to$ `APPROVED` / `REJECTED` / `REQUEST_CHANGES`).
3. **Personnel Directory & Directory Search**: Enable leaders to search coordinators and candidates by name, phone number, and jurisdiction with profile image inspection.
4. **Meeting Confirmation Workflow**: Allow coordinators to propose upcoming meeting venues and dates to administration prior to field deployment.

5. **Progressive Web Application (PWA)**: Ensure offline screen fallback, service-worker asset caching, installable home-screen experience, and mobile responsive execution without layout-induced zoom.

2.2 System Scope

- **In-Scope**:
  - Role-based access control (RBAC) enforced via Next.js Middleware and API route handlers.
  - Multi-language interface support (English, Hindi, Telugu).
  - Dark/Light aesthetic design system with smooth CSS theme switching.
  - 4-second branded animated splash screen on application startup.
  - Aggregated statistical analytics (Location coverage pie charts, monthly trend bar charts).
  - Profile customization, password modification, and image upload.
- **Out-of-Scope**:
  - Direct financial or monetary donation processing.
  - Direct SMS gateway broadcasting (internal notifications and announcement system is implemented instead).

---

3. Complete Feature Documentation

3.1 Branded 4-Second Splash Screen

- **File Reference**: `src/components/splash-screen.tsx`, `src/app/globals.css`
- **Behavior**: Upon initial application load or browser refresh, a full-screen branded splash screen displays the official WCM logo (`/icons/icon-512x512.png`), concentric pulsing rings (`splash-ring`), an animated progress bar, and ministry title. It plays for exactly 4.0 seconds, smoothly fading out at 3.4 seconds to reveal the underlying view.
- **Responsiveness**: Wrapped in `overflow-hidden` to prevent layout expansion on compact mobile viewports.

3.2 Secure Multi-Role Authentication

- **File Reference**: `src/lib/auth.ts`, `src/app/login/page.tsx`, `src/middleware.ts`
- **Behavior**: Authentication is managed by Better Auth with a custom PostgreSQL Prisma adapter. Users authenticate using their unique `username` and `password`. Passwords are verified against salted bcrypt hashes.
- **Demo Switcher**: Includes preconfigured 1-click test credentials on `/login` for rapid testing of Admin, Coordinator, State Leader, and Zone Leader roles.

3.3 Three-Part Meeting Report Engine

A meeting report is composed of three interconnected sub-sheets:

1. **Verification Sheet (`VerificationSheet`)**: Recorded before the meeting. Captures Host Convener name, church ministry, location, phone number, meeting dates, days, expected attendance, years of ministry, stage/sound/accommodation readiness, and awareness of WCM literature.
2. **Attendance Record (`AttendanceRecord`)**: Recorded during/immediately after the meeting. Captures date, location, field executer name, coordinator name, total member count, total visiting pastors count, and field remarks.
3. **After-Meeting Report (`AfterMeetingReport`)**: Recorded post-meeting. Captures overall outcome, congregation response, desire for ongoing fellowship, observed strengths, challenges, new follower count, coordinator signature, and future event opportunities.

3.4 Meeting Proposal & Confirmation System

- **File Reference**: `src/app/coordinator/confirmations/page.tsx`, `src/app/api/confirmations/route.ts`
- **Behavior**: Coordinators submit proposed dates, venue details, and estimated attendees before conducting an outreach. Administrators review the proposal and either approve it or provide written feedback.

3.5 Hierarchical Directory ("Find People")

- **File Reference**: `src/app/find-people/page.tsx`, `src/app/api/users/route.ts`
- **Behavior**: Allows authorized leaders to search personnel by candidate name or contact number. Scoped to the leader's jurisdiction (Admins search globally; State Leaders search their State; Zone Leaders search their Zone).
- **Modal Viewer**: Displays candidate history, phone call action (`tel:`), email action (`mailto:`), and high-resolution photo viewer with direct image download.

3.6 Internal Messaging & Announcement Center

- **File Reference**: `src/app/api/notifications/route.ts`, `src/components/dashboard-layout.tsx`
- **Behavior**: Administrative and Zone leaders can dispatch broadcast announcements or individual direct messages to coordinators. Unread counts are displayed with animated pulsing badges in the top navbar and mobile bottom sheet.

3.7 Statistical Analytics & KPI Dashboard

- **File Reference**: `src/app/api/analytics/monthly-summary/route.ts`, `src/components/monthly-summary.tsx`
- **Behavior**: Aggregates verified reports for any selected Month and Year. Displays total meetings conducted, total members attended, total WCM followers, a circular SVG location distribution pie chart, and a 12-month trend bar chart.

3.8 PWA & Offline Support

- **File Reference**: `public/sw.js`, `public/manifest.json`, `public/offline.html`, `src/components/pwa-provider.tsx`
- **Behavior**: Registers a service worker caching essential app icons and static bundles. When network connectivity drops, navigation requests gracefully fall back to `offline.html` styled with the WCM theme.

---

4. Technology Stack

LAYER	TECHNOLOGY	VERSION	PURPOSE
Framework	Next.js App Router	15.1.12	Core full-stack web framework & hybrid SSR/SSG
Runtime	Node.js	v20+ / v22	Server execution environment
Language	TypeScript	5.x	Strict static type checking
UI Library	React	19.0.0	Component rendering engine
Styling	Tailwind CSS	4.x	Utility-first CSS engine with `@theme` tokens
Icons	Lucide React	1.24.0	Modern SVG interface iconography
State Management	Zustand	5.0.14	Global client UI store (sidebar, language, theme)
Theme Engine	Next-Themes	0.4.6	Dark and light mode class management
Form Handling	React Hook Form	7.81.0	High-performance uncontrolled form validation
Database ORM	Prisma	5.22.0	PostgreSQL database schema modeling and client queries
Database	PostgreSQL / Supabase	15+	Relational persistence with transaction pooling
Authentication	Better Auth	1.6.23	Cookie session management & credential verification
Password Hashing	Bcrypt	6.0.0	Cryptographic password hashing (10 salt rounds)
Toast Alerts	Sonner	2.0.7	Accessible rich notification toasts

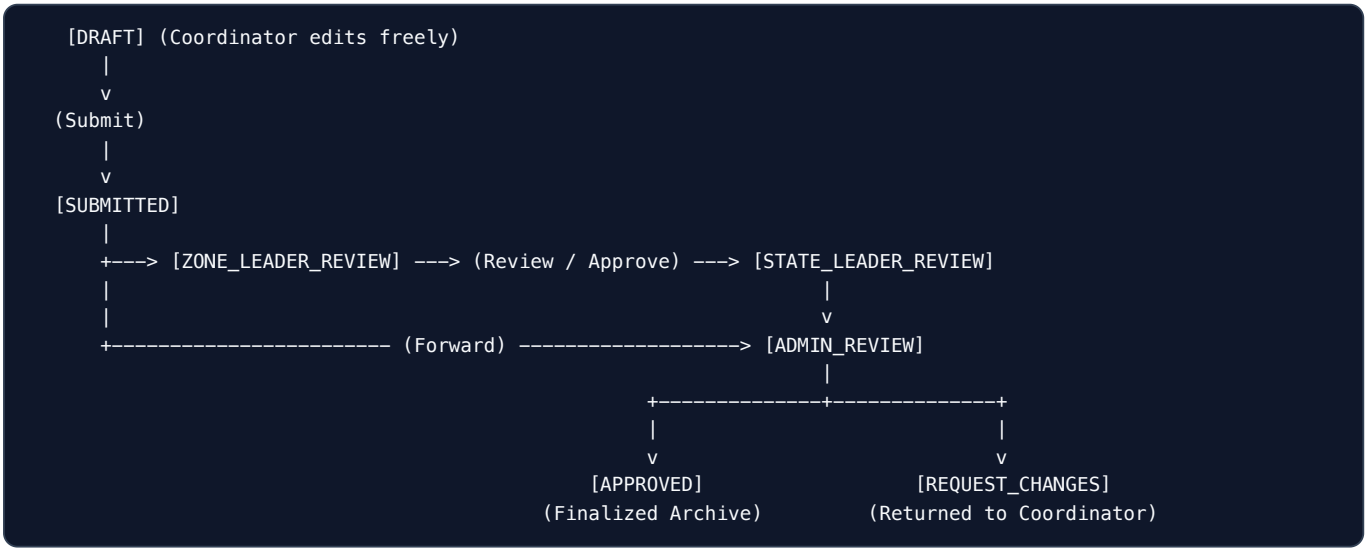
5. System Architecture

5.1 Architecture Overview

The application follows a modern **Modular Serverless Monolith** architecture built on the Next.js App Router. Client components interact with server-side API Route Handlers using standard HTTP JSON payloads. Session validation is executed at the Edge/Node boundary via `middleware.ts`.



5.2 Report Status State Machine



6. Complete Project Folder Structure

```
wcm_outreach/
├── docs/                                  # Project Technical Documentation
│   ├── Ministry_Project_Complete_Documentation.md
│   └── Ministry_Project_Complete_Documentation.pdf
├── prisma/                                # Database ORM Configuration
│   ├── schema.prisma                      # Prisma data models & relational schema
│   └── seed.ts                            # Database seeder with demo accounts
├── public/                                # Static Public Assets
│   ├── icons/                             # PWA app icons (72px to 512px)
│   ├── manifest.json                      # PWA web app manifest
│   ├── offline.html                       # Offline fallback HTML page
│   ├── sw.js                              # Custom Service Worker caching script
│   └── wcm-logo.png                       # High-resolution ministry brand logo
├── src/
│   ├── app/                               # Next.js App Router (Pages & APIs)
│   │   ├── admin/dashboard/page.tsx       # Admin control panel (Users & Reports)
│   │   ├── api/                           # Backend REST API Routes
│   │   │   ├── analytics/monthly-summary/   # Analytics aggregation endpoint
│   │   │   ├── auth/[...auth]/            # Better-Auth catch-all handler
│   │   │   ├── confirmations/             # Meeting proposal endpoints
│   │   │   ├── notifications/             # Internal messaging endpoints
│   │   │   ├── reports/                   # Report creation, submission, review
│   │   │   └── users/                     # User management & profile endpoints
│   │   ├── coordinator/                   # Coordinator Workspace
│   │   │   ├── confirmations/page.tsx     # Meeting confirmation requests
│   │   │   ├── dashboard/page.tsx         # Coordinator central dashboard
│   │   │   └── report/[id]/page.tsx       # Unified 3-step report editor wizard
│   │   ├── find-people/page.tsx           # Directory search & personnel details
│   │   ├── globals.css                    # Master CSS design system & tokens
│   │   ├── layout.tsx                     # Root layout with providers & splash screen
│   │   ├── login/page.tsx                 # Authentication page with demo selector
│   │   ├── profile/page.tsx               # Profile details & password manager
│   │   ├── state-leader/dashboard/page.tsx  # State Leader overview console
│   │   └── zone-leader/dashboard/page.tsx   # Zone Leader review console
│   ├── components/                        # Reusable React Components
│   │   ├── auth-sync-provider.tsx         # Syncs Better-Auth session with Zustand store
│   │   ├── dashboard-layout.tsx           # Responsive sidebar, navbar & mobile bottom bar
│   │   ├── monthly-summary.tsx            # Analytics charts & KPI counters
│   │   ├── pwa-provider.tsx               # PWA service worker registration & install prompts
│   │   ├── splash-screen.tsx              # 4-second animated brand startup screen
│   │   └── theme-provider.tsx             # Next-Themes dark/light provider
│   ├── lib/                               # Core Utilities & Singletons
│   │   ├── auth-client.ts                 # Better-Auth client instance
│   │   ├── auth.ts                        # Better-Auth server configuration
│   │   ├── prisma.ts                      # Global Prisma client singleton
│   │   └── translations.ts                # Localization dictionary (EN, HI, TE)
│   ├── middleware.ts                      # Route protection & role redirection middleware
│   └── store/
│       └── ui-store.ts                    # Zustand state store for navigation & theme
├── .env.example                           # Safe environment variable template
├── package.json                           # Project dependencies and build scripts
└── tsconfig.json                          # TypeScript compiler settings
```

7. User Interface Documentation

7.1 Authentication (`/login`)

- **Design Language**: Full-bleed indigo-violet gradient background, floating logo with soft radial blur, glassmorphic card container.

- **Form Controls**: Mobile-friendly 16px inputs (`text-base sm:text-sm`) to eliminate iOS Safari auto-zoom. Toggleable password visibility eye icon.
- **Demo Selector**: Quick 1-click credential buttons for Admin, Coordinator, State Leader, and Zone Leader.

7.2 Main Dashboard Frame (`src/components/dashboard-layout.tsx`)

- **Desktop Navigation**: Fixed left-hand gradient sidebar (`w-65`) with brand logo, role indicator, navigation links, and profile badge.
- **Mobile Navigation**: Hidden sidebar on small screens; replaced with an ergonomic fixed bottom navigation bar (`fixed bottom-0 left-0 right-0 z-30`) with safe-area inset padding (`pb-safe`).
- **Top App Bar**: Sticky glassmorphic navbar with theme toggle (Sun/Moon), language selector (EN, HI, TE), notification bell with unread ping indicator, and user avatar.

7.3 Coordinator Workspace (`/coordinator/dashboard`)

- **Welcome Banner**: Gradient card displaying greeting and ministry overview.
- **Action Modules**: "Create New Report" button and "Meeting Confirmations" quick link.
- **Announcements Feed**: Live broadcast messages sent by administrators.
- **Reports List**: Dual-mode rendering: responsive card list on phones (`md:hidden`) and sortable table on tablets/desktops (`hidden md:block overflow-x-auto`).

7.4 Report Wizard (`/coordinator/report/[id]`)

- **Stepped Navigation**: 3-step navigation tab bar (1. Verification, 2. Attendance, 3. After Meeting) with individual completion percentages.
- **Auto-Save & Submit**: "Save Draft" updates the database in real-time. "Submit Report" is enabled only when all mandatory fields pass validation.
- **Review Feedback Alert**: Displays reviewer notes if a report was returned with status `REQUEST_CHANGES`.

7.5 Directory ("Find People") (`/find-people`)

- **Live Search**: Debounced search input filtering across name, username, and telephone.
- **Role Filter**: Dropdown allowing filtering by role (scoped to leader authority).
- **Modal Viewer**: Candidate details modal with phone call and email triggers, jurisdiction badges, and profile image viewer.

---

8. Database Documentation

The database schema is defined in `prisma/schema.prisma` and backed by PostgreSQL.

8.1 Model: `User` (`users`)

Stores all ministry personnel credentials, jurisdiction, and profile details.

FIELD	TYPE	ATTRIBUTES	DESCRIPTION
`id`	`String`	`@id @default(uuid())`	Unique primary key
`username`	`String`	`@unique`	Login username
`passwordHash`	`String`	`@map("password_hash")`	Bcrypt hashed password
`role`	`Role`	Enum	`ADMIN`, `COORDINATOR`, `STATE_LEADER`, `ZONE_LEADER`
`name`	`String`		Full legal or ministry name
`email`	`String`	`@unique`	Contact email address
`emailVerified`	`Boolean`		Email verification flag
`phoneNumber`	`String?`		Mobile contact number
`image`	`String?`		Base64 or URL profile picture
`state`	`String?`		Assigned state jurisdiction (e.g., "AP")
`zone`	`String?`		Assigned zone jurisdiction (e.g., "Zone-1")
`part`	`String?`		Assigned territory part (e.g., "Part-1")
`createdAt`	`DateTime`	`@default(now())`	Registration timestamp
`updatedAt`	`DateTime`	`@updatedAt`	Last modification timestamp

8.2 Model: `MeetingReport` (`meeting_reports`)

The parent record for every outreach meeting report.

FIELD	TYPE	ATTRIBUTES	DESCRIPTION
`id`	`String`	`@id @default(uuid())`	Primary key
`reportId`	`String`	`@unique`	Human-readable ID (e.g. `MR-2026-00015`)
`leaderId`	`String`	Foreign key $\to$ `User.id`	Coordinator who filed the report
`status`	`ReportStatus`	`@default(DRAFT)`	Workflow status
`version`	`Int`	`@default(1)`	Report version counter
`submittedAt`	`DateTime?`		Date of submission
`createdAt`	`DateTime`	`@default(now())`	Creation date
`updatedAt`	`DateTime`	`@updatedAt`	Last updated date

8.3 Model: `VerificationSheet` (`verification_sheets`)

FIELD	TYPE	DESCRIPTION
`convenerName`	`String`	Name of host pastor or meeting convener
`churchMinistryName`	`String`	Host church ministry name
`stateDistrictCity`	`String`	Location description
`contactNumber`	`String`	Contact phone number
`meetingDates`	`String`	Scheduled meeting dates
`numberOfDays`	`Int`	Duration of outreach in days
`expectedAttendance`	`Int`	Anticipated crowd size
`yearsOfMinistry`	`Int`	Host pastor's ministry experience
`churchMembers`	`Int`	Registered church congregation size
`workersLeaders`	`Int`	Assisting volunteers count
`knowAboutWcm`	`String`	Discovery source for WCM ministry
`readWcmBooks`	`Boolean`	Familiarity with Prophet Kiran's literature
`attendedProphetMeeting`	`Boolean`	Prior attendance at WCM events
`meetingHallStage`	`String`	Stage and venue preparation readiness
`soundSystem`	`String`	Audio system readiness
`accommodationTeam`	`String`	Lodging arrangements for ministry team
`volunteerSupport`	`String`	Local volunteer readiness

8.4 Model: `AttendanceRecord` (`attendance_records`)

FIELD	TYPE	DESCRIPTION
`date`	`DateTime`	Meeting execution date
`location`	`String`	Exact venue address / church
`fieldExecuter`	`String`	Name of field executer
`coordinator`	`String`	Coordinator in charge
`membersAttended`	`Int`	Count of regular congregation members
`pastorsAttended`	`Int`	Count of visiting ministers / pastors
`remarks`	`String`	Coordinator observations & notes

8.5 Model: `AfterMeetingReport` (`after_meeting_reports`)

FIELD	TYPE	DESCRIPTION
`overallMeetingOutcome`	`String`	Evaluation of meeting impact
`congregationResponse`	`String`	Audience receptivity
`continueFellowship`	`Boolean`	Desire for continued WCM association
`futureOpportunities`	`String[]`	Array of future event opportunities
`localChurchInfluence`	`String`	Influence assessment of local church
`networkOfPastors`	`String`	Ministerial network strength
`strengthsObserved`	`String`	Key highlights and positive outcomes
`challengesObserved`	`String`	Impediments or difficulties faced
`opportunityForWcm`	`String`	Future partnership potential for WCM
`coordinatorName`	`String`	Signer name
`designation`	`String`	Ministry designation
`signature`	`String`	Textual signature
`submissionDate`	`DateTime`	Date of sign-off
`followingWcmMembersCount`	`Int`	New members connecting with WCM

---

9. API Documentation

All API endpoints are hosted under `/api/` and require valid session authentication unless explicitly public.

9.1 Authentication Endpoints

- `POST /api/auth/[...auth]`: Managed by Better Auth. Handles `signIn.email`, `signIn.username`, `signOut`, `getSession`.
  - **Headers**: `Content-Type: application/json`
  - **Cookies Issued**: `better-auth.session_token` (HTTP-only, Secure in production).

9.2 User Management Endpoints

- `GET /api/users`: List personnel. Supports search and role query parameters (`?search=john&role=COORDINATOR`).
- `POST /api/users`: Create a new coordinator or leader (Admin only).
  - **Payload**: `{ username, name, email, password, role, state?, zone?, part?, phoneNumber? }`
- `GET /api/users/[id]`: Fetch specific user profile details.
- `PUT /api/users/[id]`: Update role, jurisdiction, or reset password.
- `DELETE /api/users/[id]`: Delete a user (Cascades sessions, notifications, and reports).
- `GET /api/users/profile`: Get current logged-in user profile, stats, and recent reports.
- `PUT /api/users/profile`: Update personal name, email, phone, avatar image, or password.

9.3 Meeting Report Endpoints

- `GET /api/reports`: Returns reports scoped to caller's role (Coordinators see their own; Zone/State Leaders see their jurisdiction; Admins see all).
- `POST /api/reports/create`: Generates a new empty draft report with unique auto-incrementing ID.
- `GET /api/reports/[id]`: Fetch complete report including verification, attendance, and after-meeting sub-sheets.
- `PUT /api/reports/[id]`: Save updates to verification, attendance, or after-meeting sheets.
- `DELETE /api/reports/[id]`: Delete a draft report (Only allowed for `DRAFT` status).
- `POST /api/reports/[id]/submit`: Validate completeness and advance report status from `DRAFT` $\to$ `SUBMITTED`.
- `POST /api/reports/[id]/review`: Review action (`APPROVE`, `REQUEST_CHANGES`, `REJECT`) with required comment.
- `POST /api/reports/[id]/forward`: Forward report to `STATE_LEADER`, `ADMIN`, or `BOTH`.

9.4 Meeting Confirmations & Notifications

- `GET /api/confirmations`: Fetch proposal requests.
- `POST /api/confirmations`: Submit meeting proposal (`meetingDates`, `location`, `expectedAttendance`, `message`).
- `PUT /api/confirmations/[id]`: Admin decision endpoint (`status: "APPROVED" | "REJECTED"`, `adminFeedback`).
- `GET /api/notifications`: Get list of notifications received by user.
- `POST /api/notifications`: Send an announcement or direct message (`recipientId`, `title`, `message`).
- `PUT /api/notifications/[id]/read`: Mark specific message as read.

9.5 Analytics

- `GET /api/analytics/monthly-summary?month=10&year=2026`:
  - **Response**:

```
{
  "summary": {
    "totalMeetings": 12,
    "totalMembersAttended": 1450,
    "totalWcmFollowers": 380
  },
  "pieChartData": [
    { "name": "Vijayawada", "count": 6, "percentage": 50 },
    { "name": "Guntur", "count": 6, "percentage": 50 }
  ],
  "barChartData": [
    { "month": "Jan", "meetings": 2, "attendees": 240 },
    ...
  ]
}
```

---

10. Authentication and Security

10.1 Authentication Flow

1. **Credentials**: Client submits `username` and `password` to `/api/auth/sign-in/username`.
2. **Password Verification**: Better Auth queries `User` table; Bcrypt verifies password hash.
3. **Session Cookie**: Server generates cryptographically random token saved in `Session` table and sets `better-auth.session_token` HTTP-only cookie.

4. **Middleware Interception**: Every protected request is intercepted by `src/middleware.ts` before reaching any route handler or page component.
5. **RBAC Routing**: Non-privileged users attempting to access routes outside their role are redirected to their assigned dashboard.

10.2 Security Safeguards

- **Encrypted Passwords**: Passwords hashed with Bcrypt (10 salt rounds). Plaintext passwords are never stored.
- **HTTP-Only Cookies**: Prevents client-side XSS access to active authentication session tokens.
- **Cascading Foreign Key Integrity**: Deleting a user safely cleans up orphan sessions, accounts, and audit records.
- **Input Validation**: Backend endpoints validate mandatory fields and numeric ranges before initiating database mutations.
- **Safe Public Assets**: Service worker explicitly bypasses `/api/` calls to ensure authentication and data mutations are never stored in browser caches.

---

11. Installation and Local Setup

11.1 Prerequisites

- **Node.js**: Version 20.x or 22.x LTS installed.
- **NPM**: Version 10+ installed.
- **PostgreSQL Database**: Local PostgreSQL server or Supabase Cloud instance.
- **Git**: Installed for version control.

11.2 Step-by-Step Setup

1. **Clone Repository**:

```
git clone https://github.com/saikiranrayudu02-cell/wcm-outreach.git
cd wcm-outreach
```

2. **Install Dependencies**:

```
npm install
```

3. **Configure Environment Variables**:

```
cp .env.example .env
# Edit .env and supply your PostgreSQL connection string and secrets
```

4. **Initialize Database Schema**:

```
npx prisma generate
npx prisma db push
```

5. **Seed Demo Accounts**:

```
npx tsx prisma/seed.ts
```

6. **Start Development Server**:

```
npm run dev
# Open http://localhost:3000 in your browser
```

7. **Verify Production Build**:

```
npm run build
```

---

12. Environment Configuration

All required environment variables are listed in `.env.example`.

VARIABLE NAME	REQUIRED	PURPOSE	SAFE EXAMPLE VALUE
`DATABASE_URL`	**Yes**	PostgreSQL connection string	`postgresql://user:pass@host:5432/postgres`
`BETTER_AUTH_SECRET`	**Yes**	32-byte secret for session encryption	`openssl rand -base64 32`
`BETTER_AUTH_URL`	**Yes**	Canonical deployment root URL	`http://localhost:3000` / `https://wcm-outreach.vercel.app`
`NEXT_PUBLIC_APP_URL`	**Yes**	Frontend client base URL	`http://localhost:3000`
`NEXT_PUBLIC_SUPABASE_URL`	Optional	Supabase storage / service URL	`https://[PROJECT-REF].supabase.co`
`NEXT_PUBLIC_SUPABASE_PUBLISHABLE_KEY`	Optional	Supabase client anon key	`sb_anon_key_...`

> [!CAUTION] Never commit real `.env` or production credentials to GitHub. Always use `.env.example` as a template and configure secrets directly in the Vercel or hosting platform environment settings.

---

13. Testing and Quality Assurance

13.1 Build & Compile Testing

- **Next.js Production Compilation**: Verified via `npm run build`. Compiles all 20 static and dynamic routes cleanly with 0 TypeScript errors.
- **Linting**: Verified via `npm run lint` with Next.js ESLint configuration.

13.2 Responsive Device Emulation Matrix

Tested using headless browser viewport emulation:

- **320 × 568 (iPhone SE Compact)**: Pass (0 horizontal scroll, stacked cards).
- **375 × 812 (iPhone 11/12/Mini)**: Pass (Full bottom bar, 16px touch inputs).
- **390 × 844 (iPhone 13/14/15)**: Pass (Card layouts and modals aligned).
- **768 × 1024 (iPad Tablet)**: Pass (Sidebar drawer and expanded table views).
- **1280 × 800 (Desktop Laptop)**: Pass (Dual-pane layout with persistent sidebar).

13.3 Untested Areas & Recommendations

- Automated end-to-end Cypress or Playwright test suites have not yet been integrated into the repository.
- Unit test coverage (e.g. Jest or Vitest) should be added in future sprints for individual validation functions.

---

14. Deployment Documentation

14.1 Hosting Platform: Vercel

The application is preconfigured for continuous deployment via **Vercel**:

- **Framework Preset**: Next.js
- **Build Command**: `next build`
- **Install Command**: `npm install`
- **Node.js Version**: 20.x

14.2 Supabase IPv4 Pooler Configuration

Direct Supabase connection endpoints (`db.[REF].supabase.co:5432`) are IPv6-only. Serverless platforms like Vercel East (iad1) operate in IPv4 environments and will fail with `PrismaClientInitializationError: Can't reach database server`.

- **Required Fix**: Always configure `DATABASE_URL` using the Supabase IPv4 Pooler address:

```
postgresql://postgres.[REF]:[PASSWORD]@aws-0-[REGION].pooler.supabase.com:5432/postgres
```

14.3 Postinstall Generation

`package.json` includes `"postinstall": "prisma generate"`, ensuring the Prisma binary client is automatically compiled during the Vercel build process.

---

15. Troubleshooting Guide

ISSUE	CAUSE	SOLUTION
`PrismaClientInitializationError` (Can't reach db)	Direct IPv6 Supabase connection used on Vercel	Switch `DATABASE_URL` in Vercel to Supabase Pooler (`aws-0-*.pooler.supabase.com:5432`).
Mobile Screen Auto-Zoom on iOS	Form input font-size smaller than 16px	Enforced in `globals.css` with `@media (max-width: 768px) { input { font-size: 16px !important; } }`.
Horizontal Scrollbar on Mobile	Missing `overflow-x: clip` or flex child min-width	Ensure `min-w-0 max-w-full` on flex containers and `overflow-x: clip` on `body`.
Splash Screen Stuck	JavaScript error in third-party library	Splash screen includes a hard unmount timer at 4.0s. Check browser console.
Vercel `allowScripts` Warning	NPM 11 warning for native install scripts	Added `allowScripts` section to `package.json` for Prisma, Bcrypt, and Sharp.

16. Limitations and Future Improvements

16.1 Verified Limitations

1. **Photo Storage Format**: Currently, user profile pictures and signatures are stored as base64 data URLs directly in the database. For very large image uploads, this increases database footprint.
2. **Automated Integration Tests**: The repository lacks automated unit and E2E test scripts.

16.2 Planned Enhancements

1. **Cloud Object Storage**: Transition photo uploads from database base64 strings to Supabase Storage Buckets (`/storage/v1/object`).
2. **Push Notifications**: Integrate Web Push API for native mobile notifications when a report is approved or returned for changes.
3. **Excel / PDF Export**: Add client-side PDF export for individual meeting reports so coordinators can print hard copies for church archives.

17. User Guide

17.1 For Outreach Coordinators

1. **Log In**: Visit `/login`, enter credentials, and tap **Sign In**.
2. **Start a New Report**: On the dashboard, tap **Create New Report**.
3. **Fill Verification Sheet**: Enter host pastor details, church name, phone number, and venue readiness. Tap **Save Draft**.
4. **Fill Attendance Record**: After the outreach service, record member and pastor attendee counts.
5. **Fill After-Meeting Sheet**: Enter observations, congregation response, and textual signature.
6. **Submit**: Once all three tabs reach 100%, tap **Submit Report**.

17.2 For Zone & State Leaders

1. **Review Submissions**: Access `/zone-leader/dashboard` or `/state-leader/dashboard`.
2. **Inspect Reports**: Tap **View Details** or **Edit** on any report under review.
3. **Approve or Request Changes**: Enter review remarks. Tap **Approve** to forward or **Request Changes** to return to the coordinator.
4. **View Monthly Summary**: Switch to the **Analytics** tab to view attendance charts and location distribution.

17.3 For System Administrators

1. **User Management**: In `/admin/dashboard`, tap **Create User** to register coordinators and leaders.
2. **Announcements**: Switch to **Send Messages** tab to broadcast ministry updates.
3. **Confirmations**: Review upcoming outreach proposals and provide administrative feedback.

---

18. Developer Maintenance Guide

18.1 Adding a New Field to Reports

1. Update `prisma/schema.prisma` under the relevant model (`VerificationSheet`, `AttendanceRecord`, or `AfterMeetingReport`).
2. Run database migration:

```
npx prisma db push
npx prisma generate
```

3. Update the form state and inputs in `src/app/coordinator/report/[id]/page.tsx`.
4. Update the review view in `src/app/coordinator/dashboard/page.tsx` and admin dashboards.

18.2 Adding Translations

1. Open `src/lib/translations.ts`.
2. Add your English key to the `translations` object, providing corresponding strings for `hi` (Hindi) and `te` (Telugu).
3. Use `translate("Your Key", language)` inside any React component.

---

19. Glossary and References

19.1 Glossary

- **Convener**: The local host pastor or minister organizing the church venue for the WCM outreach meeting.
- **Field Executer**: The designated minister delivering the sermon or executing ministry on the field.
- **PWA (Progressive Web App)**: Web application technology allowing desktop and mobile users to install the portal to their home screen and access offline cached resources.
- **RBAC (Role-Based Access Control)**: Security authorization restricting route access based on user role (`ADMIN`, `STATE_LEADER`, `ZONE_LEADER`, `COORDINATOR`).
- **Supabase Pooler**: An IPv4 transaction and session connection proxy for PostgreSQL databases operating in serverless cloud environments.

19.2 References

- **Next.js Documentation**: nextjs.org/docs
- **Prisma ORM Documentation**: prisma.io/docs
- **Better Auth Documentation**: better-auth.com
- **Tailwind CSS v4 Documentation**: tailwindcss.com/docs
- **Supabase Connection Pooling**: supabase.com/docs/guides/database/connecting-to-postgres#connection-pooler

---

End of Documentation — World Christian Ministry (WCM)